Notes 4/7: The Rationals

Def 1: Let $Q = \{(x,y) : x,y \in \mathbb{Z} \text{ and } y \neq 0\}$

we call x the numerator and y the denominator

(x,y) is a numerator denominator pair : $\frac{x}{y}$

Note: $\frac{1}{2} = \frac{3}{6}$ in $\mathbb{Q}$ however $(1,2) \neq (3,6)$

Def 2: (x,y) and (z,w) in Q are defined to be rationally equivalent if $\cancel{xz = yw}$. In this case we write $\cancel{(x,y) \sim (z,w)}$ $(x,y) \sim (z,w)$

$\cancel{(x,w) \sim (y,z)}$

$xw = yz$

Thm 1. $\sim$ is an equivalence relation

reflexive

$(x,y) \sim (x,y)$

Symmetric

$(x,y) \sim (z,w) \Rightarrow (z,w) \sim (x,y)$

Transitive

if $(x,y) \sim (z,w)$, $(z,w) \sim (v,s)$, then $(x,y) \sim (v,s)$

Def 3. We write $[x,y]$ to denote the equivalence class containing (x,y)

I.E., $[x,y] = \{(a,b) \in Q : (a,b) \sim (x,y)\} = \{(a,b) \in Q : ay = bx\}$

Ex. $[1,2] = \{(a,b) \in Q : (a,b) \sim (1,2)$

$= \{(a,b) \in Q ; 2 \cdot a = 1 \cdot b\}$

$[-4,-6] = \{(a,b) \in Q : -6a = -4b$

$-2, 3a = -2 \cdot 2b$

$3a = 2b$

Def 4: Define the rational #s to be $\mathbb{Q} \overset{def}{=} \{[x,y] : (x,y) \in Q\}$

$= \{[x,y] : x,y \in \mathbb{Z}, y \neq 0\}$

Thm 2: Let $[x,y] \in \mathbb{Q}$. Then $[x,y] = [0,1]$ if and only if $x=0$

Lemma: $[x,y] = [a,b] \iff xb = ay$

Proof sketch: If $[x,y] = [a,b]$, then $(x,y) \in [a,b]$ since $(x,y) \in [x,y]$

Thus $(x,y) \sim (a,b)$ so $xb = ay$

If $xb = ay$, then $(x,y) \sim (a,b)$, so $(x,y) \in [a,b]$. If $(z,w) \in [x,y]$

then $(z,w) \sim (x,y)$. By transitivity $(z,w) \sim (a,b)$, so $(z,w) \in [a,b]$

Thus $[x,y] \subset [a,b]$

Proof of Thm 2: If $[x,y] = [0,1]$ then $x \cdot 1 = y \cdot 0$ by lemma.

conversely, if $x=0$, then $x \cdot 1 = 0 = y \cdot 0$, Thus $[x,y] = [0,1]$ by Lemma.

Thm 3 $[x,y] = [1,1] \iff x = y$

Def 5. Define the binary operations $+$ and $\cdot$ on $\mathbb{Q}$ as follows:

If $[x,y]$ and $[a,b]$ are in $\mathbb{Q}$ $[x,y] \cdot [a,b] \overset{def}{=} [x \cdot a, y \cdot b]$

$[x,y] + [a,b] \overset{def}{=} [xb + ay, yb]$

$$\frac{x}{y} + \frac{a}{b} = \frac{xb}{yb} + \frac{ay}{yb} = \frac{xb+ay}{yb}$$

Ex. $[1,2] = [3,6]$ and $[2,3] = [-4,-6]$ so we need to have

$[1,2] + [2,3] = [3,6] + [-4,-6]$ and $[1,2] \cdot [2,3] = [3,6] \cdot [-4,-6]$

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$[7,6] \quad\quad [-42,-36] \quad\quad [2,6] \quad\quad [-12,-36]$

$6(-42) = 7(-36) \quad\quad 2(-36) = 6 \cdot (-12)$

lemmas 4 and 5 : $+, \cdot$ are well defined

Thm 6. the additive identity is $[0,1]$
the multiplicative identity is $[1,1]$

Thm 7. $[x,y] = [cx, cy]$ if $c \in \mathbb{Z}$ and $c \neq 0$.

Thms 8, 9, 10. $+, \cdot$ are commutative and $[x,y] \cdot [y,x] = [1,1]$ if $x,y \neq 0$

Thm 11. ($\mathbb{Q}$ is a field) Every element of $\mathbb{Q}$ other than $[0,1]$ has a multiplicative inverse. $(\frac{3}{2} \cdot \frac{2}{3} = 1)$

Proof. let $[x,y] = \mathbb{Q}$ and suppose $[x,y] \neq [0,1]$
Thus $x \neq 0$. Hence $[y,x] \in \mathbb{Q}$. We compute $[x,y] \cdot [y,x] = [xy, xy]$
$= [1,1]$

Def 6. The canonical embedding $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is defined by $a \mapsto [a,1]$ for every $a \in \mathbb{Z}$

Thm 12. the canonical embedding is injective

Proof. Suppose $a \mapsto [c,1]$ and $b \mapsto [c,1]$
Thus $[a,1] = [c,1]$ and $[b,1] = [c,1]$. Hence $[a,1] = [b,1]$
so $a \cdot 1 = b \cdot 1$ or $a = b$

$+, \cdot, -$ all behave well with respect to embedding
$1+2 = 3 \mapsto [3,1]$
$[1,1] + [2,1] = [1 \cdot 1 + 1 \cdot 2, 1] = [3,1]$